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Improving predictive performance for molecular ADMET properties using a chemical language model

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Abstract

Molecular encoders play a central role in AI-driven drug design by defining the chemical representations used for downstream prediction and decision-making. However, many widely used encoders are not explicitly trained to internalize ADMET-relevant structure–property relationships across diverse endpoints. In this study, we fine-tuned a DeBERTa-based SMILES (Simplified Molecular Input Line Entry System) encoder to internalize ADMET-relevant structure–property information across 22 endpoints under a multi-task learning (MTL) framework, while maintaining strong comprehension of SMILES syntax and structural regularities. Starting from a pretrained ZINC-based DeBERTa checkpoint, we trained on a 300 K PubChem–ADMET dataset using a multi-output regression scheme with a Focal MAE objective to mitigate optimization imbalance across heterogeneous endpoints. To consistently assess balanced performance across tasks with heterogeneous metrics, we further introduced the ADMET-Balanced Performance Score (ABPS) as an integrated evaluation criterion. On the TDC benchmark leaderboard, our encoder achieved state-of-the-art performance on 12 endpoints and top-5 performance on 7 endpoints, indicating broad competitiveness across ADMET tasks. In addition, by restoring an MLM head and tracking MLM accuracy throughout fine-tuning, we show that DeBERTa preserves SMILES token-level syntax competence more stably than BERT- and RoBERTa-based encoders under the same ADMET training conditions. Overall, this work establishes an ADMET-aware SMILES encoder that balances broad-spectrum endpoint learning with preservation of pretrained SMILES knowledge, and can serve as a foundation for downstream molecular modeling pipelines, including future multimodal drug-design systems.

KEYWORDS

ADMET, AI-driven drug design, DeBERTa, SMILES encoder

INTRODUCTION

Multi-modal learning has achieved strong performance across many domains by fusing complementary information from different data types (e.g., text, images, and molecular structures).[1–4] In such systems, overall performance is often bounded by the quality of the *pretrained encoders* that convert each modality into informative representations suitable for downstream fusion.[5] In drug design in particular, the molecular encoder plays an outsized role: it must translate chemical information—commonly represented as SMILES (Simplified Molecular Input Line Entry System) strings[6]—into embeddings that remain faithful to molecular structure while being directly useful for pharmacologically meaningful decision-making. Accordingly, rather than studying multi-modal fusion strategies themselves, this work targets the development of a high-quality pretrained SMILES encoder intended to serve as a stronger and more discriminative backbone for multi-modal drug design models.

To achieve the goal of developing such high-quality encoders, recent research has increasingly focused on adapting sophisticated Transformer-based architectures, such as BERT and RoBERTa, to the field of molecular representation. For instance, Smile-to-Bert enhanced the pharmacological relevance of embeddings by multitasking the prediction of physical descriptors directly from SMILES strings.[7] Similarly, MolEncoder optimized masked language modeling strategies through sophisticated noise-injection and masking ratios to maximize the information density of the resulting molecular representations.[8] Furthermore, MolRoPE-BERT integrated rotary position embeddings to better capture the complex, long-range sequential dependencies inherent in chemical structures.[9] The choice of molecular representation itself has been shown to significantly affect deep learning model performance, suggesting that SMILES-based approaches may offer certain advantages in capturing sequential structural features.[10] While these works collectively demonstrate the increasing capability of pretrained SMILES encoder to extract high-dimensional features from raw SMILES data, this structural proficiency does not inherently translate into a nuanced understanding of biological outcomes.

A central practical challenge is that many existing SMILES encoders excel at learning chemical "language" regularities—token syntax, frequent substructures, and local patterns—yet remain comparatively weak at capturing ADMET (Absorption, Distribution, Metabolism, Excretion, and Toxicity), which often determines whether a candidate molecule succeeds or fails in real-world development.[11] Even when an encoder produces chemically plausible representations, it may overlook subtle pharmacokinetic liabilities or toxicophores that drive poor bioavailability, adverse metabolism, or toxicity. Because ADMET-related issues are a leading cause of late-stage attrition, inadequate ADMET awareness at the representation level can directly undermine both screening and generation, limiting the practical utility of AI-driven molecular design.

To address this gap, we propose an encoder-centric approach that injects deep ADMET knowledge while preserving foundational chemical understanding. Concretely, we finetune a pretrained SMILES encoder to emphasize representations predictive of toxicity risk, metabolic behavior, and other ADMET-relevant signals, while explicitly guarding against catastrophic forgetting of chemical grammar and structural priors. Importantly, this is not a routine finetuning procedure: we incorporate a knowledge-preservation objective by monitoring masked language modeling (MLM) accuracy during adaptation, ensuring that improvements in ADMET sensitivity do not come at the cost of degraded structural understanding. Moreover, we explicitly position the resulting model as an ADMET-aware SMILES encoder trained under a multi-task learning (MTL) framework, allowing it to learn shared, cross-endpoint pharmacological representations rather than optimizing for a single property in isolation. This MTL design serves as a key methodological contribution of our approach and strengthens the encoder's ability to generalize across diverse ADMET endpoints.

Our design leverages the architectural strengths of DeBERTa (Decoding-enhanced BERT with Disentangled Attention),[12] whose disentangled attention and relative positional encoding are well aligned with the order-sensitive and syntax-constrained nature of SMILES sequences. The resulting DeBERTa-based SMILES encoder is thus characterized by (i) robust structural comprehension as reflected in retained MLM performance, and (ii) enhanced ADMET-oriented predictive accuracy—together forming an advanced feature extractor intended to improve the practical utility and predictive reliability of downstream drug design pipelines.

BACKGROUND

DeBERTa extends BERT/RoBERTa by introducing (i) a disentangled attention mechanism that explicitly separates token content from token position information and (ii) an Enhanced Mask Decoder (EMD) that reinjects absolute position signals at the decoding stage. In conventional BERT/RoBERTa-style encoders, absolute positional embeddings are added to each token embedding[13,14] so that content and position are fused into a single vector representation for every token. In contrast, DeBERTa represents each token using two distinct vectors—one for content and one for (relative) position—and computes attention by combining these signals in a structured, factorized way.

Specifically, for tokens at positions i and j, the attention compatibility score can be decomposed into four components:

$$A_{i,j} = \{H_i, P_{i|j}\} \times \{H_j, P_{j|i}\}^T = H_i H_j^T + H_i P_{j|i}^T + P_{i|j} H_j^T + P_{i|j} P_{j|i}^T.$$

Here, H_i denotes the content vector of token i, and $P_{i|j}$ denotes the relative position vector of token i with respect to token j. Because this formulation relies on relative position embeddings, the position-to-position term $P_{i|j}P_{j|i}^T$ is empirically less informative; accordingly, DeBERTa computes attention using only the first three terms (content–content, content–position, and position–content). This design provides a direct mechanism for distance-sensitive interactions: the same pair of token contents can receive different attention depending on their relative offset and direction.

After encoding relative positional relationships throughout the Transformer layers, DeBERTa applies EMD to complement relative signals with absolute position information at the decoding stage. Rather than injecting absolute positions in the input layer (as in BERT/RoBERTa), EMD adds absolute position embeddings immediately before the softmax layer used for masked-token prediction. Mechanistically, this separation helps preserve relative-position modeling in the encoder while still enabling the model to distinguish syntactic or role-like

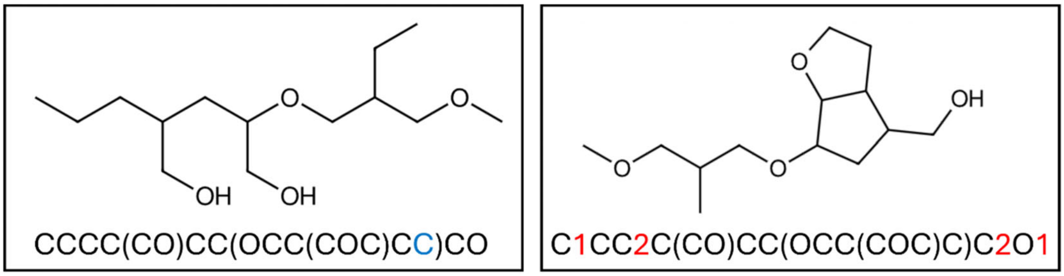


FIGURE 1 Representative molecules and their corresponding SMILES strings, illustrating how small token-level edits can yield substantial changes in molecular structure. Inserted tokens are highlighted in red, and deleted tokens are highlighted in blue.

differences that depend on absolute location. In addition, because content and position are explicitly factorized in attention, this architecture can reduce undesirable entanglement between "what" a token is and "where" it appears.

This design can make fine-tuning more robust to newly introduced objectives by helping preserve the model's previously learned ability to parse SMILES syntax—that is, its MLM-based competence as a SMILES grammar/structure parser—under task-specific optimization. In BERT/RoBERTa, content and absolute positional information are fused at the input stage into a single token vector. As a result, optimization signals (gradients) induced by downstream task learning can simultaneously perturb the entire fused representation, so that updates driven by semantic objectives may directly distort positional encoding as well. Such coupled interference can manifest as a global representational shift, increasing the likelihood of degrading the model's learned understanding of SMILES structure. In contrast, DeBERTa is designed so that content and position information interact through separated, factorized routes during attention computation. This factorized interaction structure offers architectural flexibility that can relatively reduce the chance that semantic updates overwrite positional representations in a large-scale, entangled manner, thereby lowering the risk of catastrophic degradation in the model's ability to preserve SMILES grammatical regularities and token-to-token structural dependencies.

The implications of this structural factorization become clearer when considering the characteristics of SMILES representations. While SMILES is a linear string, it is not ordinary text: it is a domain-specific language whose token order directly specifies a molecular graph. SMILES uses a constrained token set (e.g., atom symbols, bond markers, parentheses, and ring-closure digits), where adjacent tokens typically imply bonded atoms, parentheses indicate branching, and paired numeric labels denote cyclic connections. Consequently, small perturbations in the token sequence can alter local bonding patterns, change branch attachment points, or rewire ring closures, yielding drastically different chemical graphs. Figure 1 illustrates this sensitivity, where inserting or deleting only a few tokens transforms the implied molecular structure.

Given DeBERTa's emphasis on relative-position-aware attention and SMILES' heavy dependence on token order for molecular connectivity, the two approaches are naturally complementary. Disentangled attention can model localized dependencies (e.g., adjacent atoms connected by a bond) while also capturing longer-range, nonlocal relationships (e.g., ring closures expressed by paired numeric tokens and distant branch reconvergence). By leveraging this architectural–representational synergy, a DeBERTa-based Transformer can more faithfully interpret SMILES as a structured chemical language, improving its ability to encode molecular connectivity without sacrificing the expressive power and contextual depth that characterize modern encoder-only Transformers.

METHODOLOGY

Base model

We initialized our SMILES encoder from the sagawa/ZINC-deberta checkpoint,[15] which was built by fine-tuning Microsoft's DeBERTa-base[12] on approximately 23 million canonical SMILES from the ZINC-canonicalized dataset.[16] Since our study focuses on molecular sequences rather than natural-language text, we start from this chemistry-adapted encoder and further fine-tune it for ADMET-aware representation learning. This initialization provides a strong inductive bias toward valid SMILES syntax and chemically meaningful token interactions, allowing subsequent ADMET-oriented training to focus on capturing property-relevant variations in molecular structure rather than re-learning basic SMILES "language" regularities.

Dataset

Dataset construction plays a central role in AI-driven chemistry, as the formulation of the learning problem is itself shaped by how training data is curated and annotated.[17] For ADMET-oriented training, we curated 300 thousand canonical SMILES from PubChem[18] and annotated each molecule with 22 ADMET endpoints using the ADMET-AI platform (version 1.3.1).[19] The reliability of

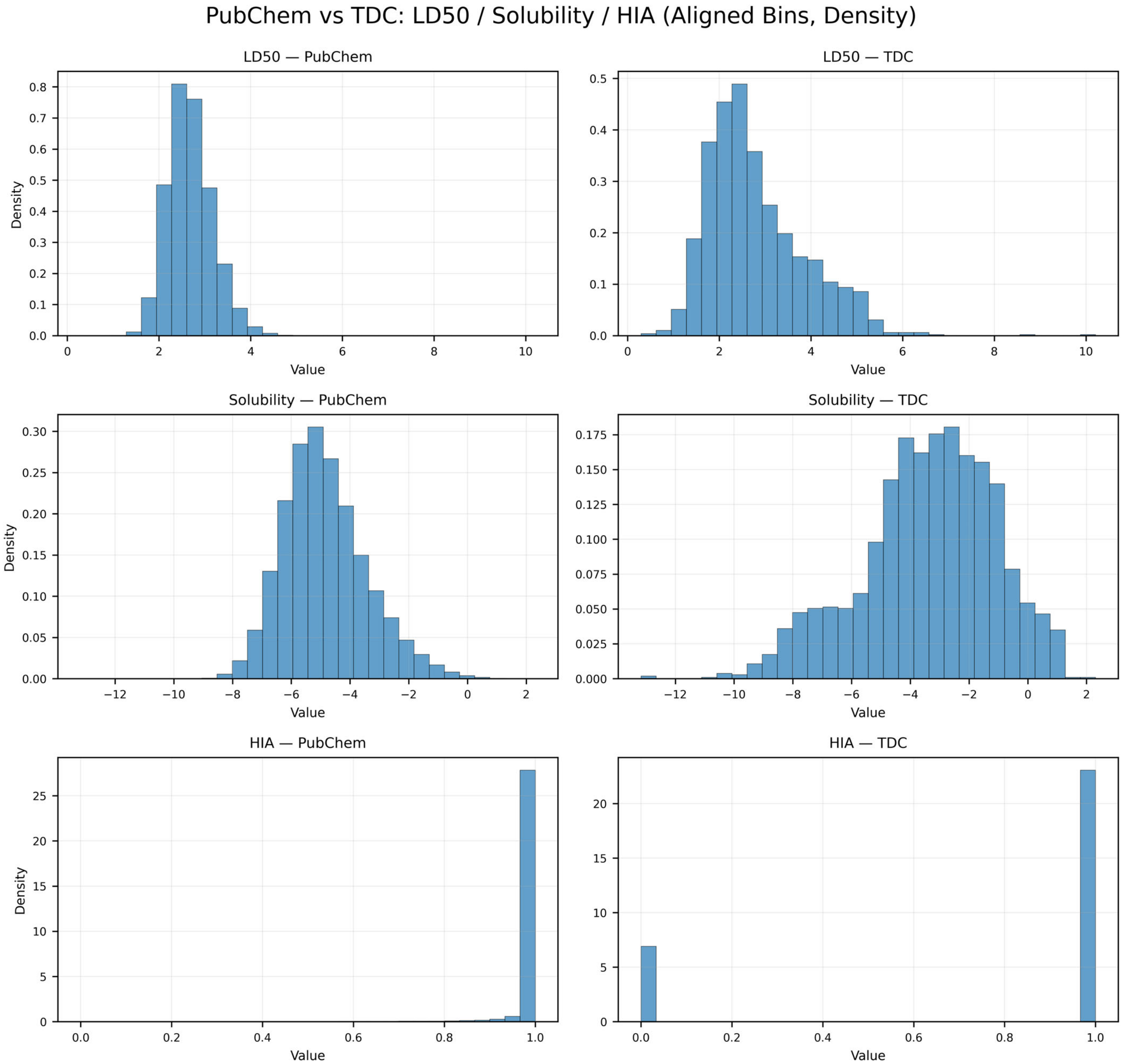


FIGURE 2 Comparison of endpoint value distributions between the PubChem–ADMET training dataset and the corresponding TDC benchmark datasets for three endpoints (LD50, solubility, and HIA) showing pronounced distribution shift and/or label imbalance.

ADMET-AI as a source of surrogate labels is supported by its extensive validation against experimental ground-truth data from the Therapeutics Data Commons (TDC). According to the benchmark results for 41 TDC datasets, the platform demonstrates high predictive accuracy, achieving an $R^2 > 0.6$ for 50% of regression tasks and an AUROC >0.85 for 65% of classification tasks.

Leveraging the high performance of the ADMET-AI platform, we treat the resulting dataset as a comprehensive reference resource for structural internalization, even though the labels are generated via a predictive system rather than direct experimental measurement. Within this framework, our encoder is trained to internalize ADMET-relevant structure–property signals at scale. This approach can introduce biases inherited from the surrogate labels (e.g., systematic errors, endpoint-specific calibration issues, or coverage gaps). We therefore explicitly acknowledge the possibility that such bias may influence downstream behavior, while also arguing that its practical impact is mitigated by (i) the chemical diversity of PubChem-scale sampling, (ii) stringent scaffold-based evaluation splits that prevent trivial memorization of chemotypes, and (iii) external benchmarking on datasets beyond the ADMET-AI-generated labels.

To characterize the applicability domain of our training distribution, we analyzed the endpoint-wise value distributions of the PubChem–ADMET dataset and compared them with the distributions encountered in TDC benchmark tasks. Histograms and coverage analyses are provided in the Supplementary Material (Figure 2 shows HIA, LD50, and Solubility only; the distributions for all endpoints are reported in Figure S1). While TDC exhibits discretized distributions for certain binary classification endpoints (e.g., CYP3A4, BBB, and AMES), we processed the PubChem–ADMET labels to retain continuous probability scores to prevent the loss of nuanced structure–property signals. This approach is consistent with the standard TDC evaluation protocol, which utilizes real-valued predicted scores to calculate binary classification performance metrics. With these high-resolution continuous signals, PubChem-derived samples cover the majority of value ranges observed across TDC endpoints, supporting the use of PubChem–ADMET as a broad pretraining/fine-tuning substrate for ADMET-aware encoders. Two notable exceptions were LD50 and Solubility, for which the PubChem–ADMET distribution showed limited coverage relative to TDC. Interestingly, for HIA we observed a markedly imbalanced distribution in PubChem–ADMET, yet the model still achieved strong performance on the corresponding TDC benchmark, suggesting that the learned representation retains robustness even under label-imbalance in the surrogate-labeled training set.

The 22 ADMET endpoints considered in this study each have distinct physical meanings and units, and consequently exhibit different numerical ranges. Such scale disparities can cause optimization imbalance in multi-task learning (MTL): the loss from certain endpoints may dominate the training signal, while endpoints with smaller ranges may be under-optimized and insufficiently learned. To mitigate this issue, we applied z-score normalization to each endpoint so that it has zero mean and unit standard deviation. This is a widely adopted preprocessing practice for balancing target variables with heterogeneous scales in multi-output regression and MTL settings.[20] By standardizing the endpoints, we ensured that each task contributes to optimization on a comparable numerical scale, improving training stability and loss balance. During evaluation, predictions were inverse-transformed to the original scale to compute performance in the native units.

We conducted a duplicate analysis based on canonical SMILES matching between the 300 000 molecule PubChem dataset and the TDC benchmark datasets to proactively check for any direct structural overlap between the two. We adopted a 5-fold scaffold split to more rigorously assess generalization to unseen chemotypes. Specifically, molecules were first grouped by Bemis–Murcko scaffolds, and scaffolds groups were then assigned to one of five folds using a greedy bin-packing strategy to balance the number of molecules per fold. For each fold, four scaffold folds were used for training and the remaining fold was used for validation. This design reduces information leakage arising from close structural analogs across splits and better reflects realistic deployment scenarios in which new candidate molecules often belong to scaffolds absent from the training set.

Finetuning

The model architecture follows a modular design consisting of a DeBERTa-base encoder and specialized task heads. The encoder comprises 12 Transformer layers with a hidden size (d_{model}) of 768 and 12 attention heads, utilizing a disentangled self-attention mechanism where content and relative position information are computed separately. Each layer incorporates an intermediate feed-forward network with a dimensionality of 3072 and a Gaussian Error Linear Unit (GELU) activation function. To verify and monitor the encoder's pretrained chemical knowledge within our data context, a Masked Language Modeling (MLM) head was integrated atop the encoder. This head consists of a transformation block—comprising a 768-unit dense layer, GELU activation, and Layer Normalization ($\epsilon = 1e^{-7}$)—followed by a final decoding layer that maps the hidden states back to the tokenized SMILES space. For the ADMET property prediction, a custom regression head was attached to the encoder's output, implemented as a Multi-Layer Perceptron (MLP) featuring an initial 768-to-128 dense layer, ReLU activation, and a dropout layer ($p = 0.3$), followed by a final linear projection to the 22 target endpoints.

To transform a pretrained model that already captures the chemical context of SMILES into an ADMET-aware SMILES encoder, we performed a fine-tuning procedure that jointly learns 22 distinct ADMET properties. Before initiating ADMET fine-tuning, we first established a protocol to verify that the pretrained encoder's ability to represent SMILES syntax and chemical "grammar" remained stable. Because the released pretrained checkpoint contains only the encoder weights—and the MLM head is not preserved and is therefore absent or randomly initialized—we froze the encoder and trained an MLM head separately. Specifically, we randomly masked 15% of SMILES tokens and optimized the MLM head using cross-entropy loss. This procedure yielded an MLM token-recovery accuracy of approximately 96%, confirming that the encoder robustly retains structural and syntactic patterns in SMILES. We then froze the MLM head and conducted multi-task fine-tuning to simultaneously predict all 22 ADMET endpoints. Throughout training, we continuously tracked MLM accuracy to quantitatively assess how optimizing the ADMET objective affected the encoder's SMILES-grammar competence (Figure 3).

For ADMET-aware fine-tuning, we adopted a multi-task learning (MTL) framework that predicts 22 endpoints concurrently. Rather than training separate models for each endpoint, we used a single shared SMILES encoder backbone and attached a multi-output regression head that produces a 22-dimensional prediction vector in a single forward pass. We extracted the encoder's final hidden

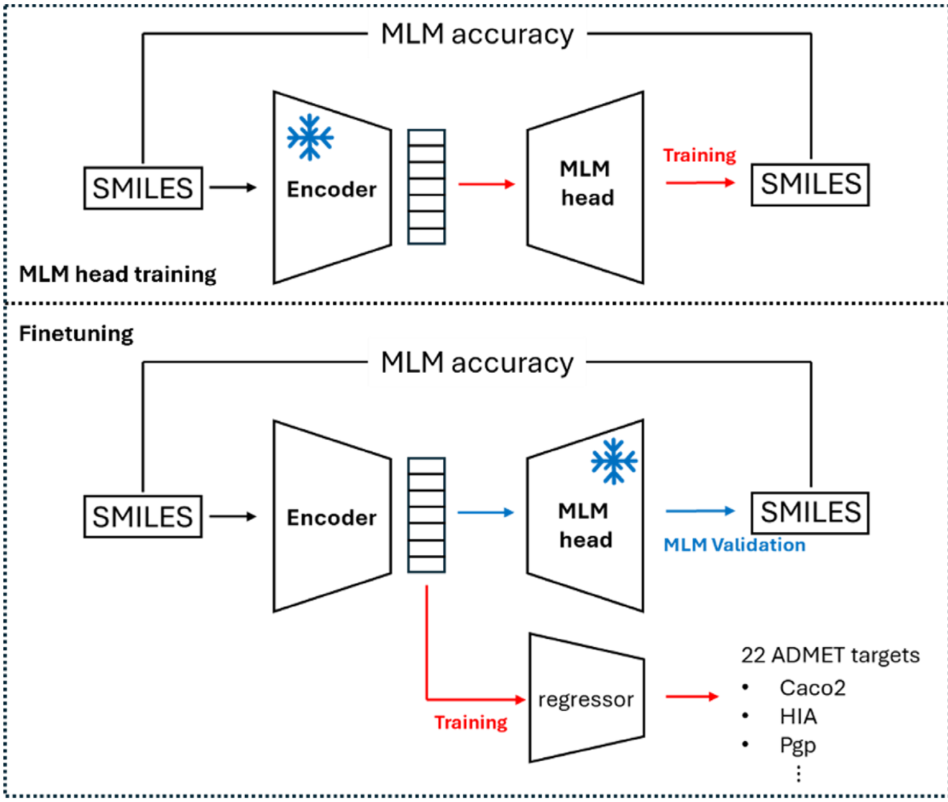


FIGURE 3 Overview of the two-stage fine-tuning protocol: Restoration of the masked language modeling (MLM) head with the encoder frozen, followed by multi-task ADMET fine-tuning while tracking MLM accuracy to monitor retention of pretrained SMILES knowledge.

states, masked padding tokens, and applied mean pooling to obtain a fixed-length, molecule-level embedding. This embedding was passed into a shallow fully connected regression head with ReLU activation and dropout regularization (0.3), followed by a final linear output layer without an additional activation function.

Optimization was performed using the AdamW optimizer, with mean absolute error (MAE) as the base loss. However, in multi-task regression settings, error distributions can be highly heterogeneous across endpoints and samples, and optimization may become dominated by specific endpoints or high-error samples. In such cases, the shared encoder's representation space can be abruptly reconfigured by a subset of loss terms, potentially perturbing previously learned structural representations.

To mitigate this issue, we introduced a Focal MAE objective that modulates each sample's contribution based on its error magnitude[21]:

$$L = \sum_i \left(1 - e^{-\gamma\left|y_i - \widehat{y}_i\right|}\right)\left|y_i - \widehat{y}_i\right|,$$

where $|y_i - \widehat{y}_i|$ is MAE loss and γ is hyper-parameter of focal loss. By applying an error-dependent scaling controlled by γ, Focal MAE encourages learning from harder samples while preventing extreme gradients from excessively distorting the representation space. This design enables more effective learning of difficult cases while maintaining representational stability during multi-endpoint training.

Prior to full ADMET multi-task training, we conducted a preliminary learning-rate study to identify a setting that supports stable convergence and representation preservation during fine-tuning. We compared four candidate learning rates—1×10^{-5}, 3×10^{-5}, 5×10^{-5}, and 7×10^{-5}—while fixing the Focal MAE decay parameter to $\gamma = 2.0$, following a commonly used default configuration.[21] At higher learning rates (5×10^{-5} and 7×10^{-5}), we observed sharp drops and unstable oscillations in MLM accuracy during early or mid training (Figure S2), indicating that overly strong optimization signals can induce large shifts in the encoder representation space and transiently disrupt the chemical-grammar features established during pretraining. Based on the most stable decrease in validation loss and the most stable MLM-accuracy trajectory, we selected 3×10^{-5} as the final learning rate for fine-tuning.

Evaluation

We evaluated the fine-tuned model's performance on the TDC (Therapeutics Data Commons) benchmark leaderboard[22] by predicting each ADMET endpoint and comparing our results with benchmark values reported by existing baselines. To ensure a rigorous and fair comparison of predictive power, we followed the TDC benchmarking protocol by utilizing the official scaffold-based data splits. While our model was trained on an expanded dataset using surrogate labels to enhance representation learning, all performance metrics were calculated on these official test sets. This ensures that our results are directly comparable to the state-of-the-art (SOTA) values on the TDC leaderboard, even though the training scales differ.

Depending on the property, we used ROC-AUC or PR-AUC for binary classification tasks, and MAE or the Spearman correlation coefficient for regression tasks, following the evaluation protocol of the TDC leaderboard. However, simply listing the per-endpoint scores across 22 tasks is insufficient to capture the central goal of this study—namely, developing a SMILES encoder that exhibits a balanced and comprehensive understanding of ADMET, rather than maximizing performance on a small subset of endpoints. Even if a model performs strongly on several tasks, markedly poor performance on other endpoints would make it difficult to interpret the representation as robust and broadly ADMET-aware.

Moreover, the TDC benchmark uses heterogeneous metrics across endpoints (ROC-AUC, PR-AUC, MAE, Spearman, etc.), which differ in scale, value range, and directionality (some are "higher is better," while others are "lower is better"). As a result, naïvely averaging raw scores or comparing them directly across endpoints is not principled. To address this, we introduced a Relative Performance Percentile (RPP) that reflects the model's relative position on the leaderboard for each endpoint, rather than its absolute score. For each endpoint i, RPP is defined as:

$$\mathrm{RPP} = \frac{S_{Model} - S_{SOTA}}{S_{ref} - S_{SOTA}},$$

where S_{Model} is the score of our model, S_{SOTA} is the score of the best-performing (state-of-the-art) leaderboard model, and S_{ref} is the score of a reference baseline corresponding to the lowest-ranked model among those reported. Under this definition, $\mathrm{RPP}_i \leq 0$ indicates performance at or above the leaderboard SOTA level, whereas $\mathrm{RPP}_i \geq 1$ indicates performance at or below the reference baseline.

To build an integrated score that aligns with our objective—balanced ADMET understanding rather than isolated peak performance—we require an aggregation scheme in which underperformance on certain endpoints meaningfully influences the overall assessment. If weaknesses on a subset of endpoints are not adequately reflected, the integrated score would fail to represent the desired notion of robust, broad-spectrum ADMET representation learning.

Accordingly, we adopted the harmonic mean as the aggregation operator. Unlike the arithmetic mean, the harmonic mean places greater weight on smaller values, ensuring that degraded performance on any endpoint produces a clear penalty in the overall score. This prevents exceptional performance on a few endpoints from compensating for poor performance on others, which matches the evaluation objective of requiring consistently strong performance across ADMET tasks.

Nevertheless, the raw RPP values can span from negative values to values greater than 1, and they are oriented such that smaller is better. Applying such values directly in a harmonic mean would be problematic because (i) the lack of boundedness allows extreme values to dominate the aggregate, and (ii) the "smaller is better" direction is not compatible with the harmonic mean's typical interpretation as a "larger is better" score. To resolve this, we applied a tanh normalization that both bounds the values to [0,1] and flips the directionality. Specifically, for each endpoint we transformed RPP into a normalized endpoint score P_i as:

$$P_i = \frac{1 - \tanh(RPP_i - 1)}{2}.$$

This transformation maps SOTA-level or better performance ($\mathrm{RPP}_i \leq 0$) to values close to 1, and baseline-level or worse performance ($\mathrm{RPP}_i \geq 1$) to values approaching 0. In addition, the saturation behavior of tanh prevents excessively high performance far above SOTA on a single endpoint from inflating the overall score disproportionately, thereby reducing the risk that one endpoint dominates the integrated metric.

Finally, we define the ADMET-Balanced Performance Score (ABPS) as the harmonic mean of the normalized endpoint scores P_i:

$$\mathrm{ABPS} = \frac{n}{\sum_{i=1}^{n} \frac{1}{P_i}},$$

where n is the number of endpoints included in the evaluation. Because the harmonic mean strongly penalizes small values, reduced performance on any endpoint is directly reflected in ABPS. At the same time, the tanh-based bounded normalization suppresses abnormal score inflation caused by extreme outliers, yielding a numerically stable integrated metric. Overall, ABPS is a purpose-driven metric designed to evaluate whether a model has learned a balanced ADMET representation, providing a more consistent view of generalization than per-endpoint score listing alone.

To verify that ABPS consistently tracks improvements in generalization during training, we analyzed the relationship between ABPS and the validation loss at each epoch. Figure 4 shows a strong correlation between the two: as validation loss decreases, ABPS increases

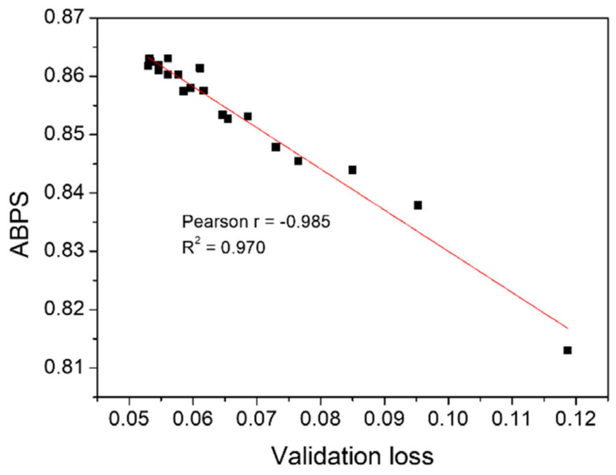


FIGURE 4 Relationship between validation loss and the proposed integrated metric (ABPS) across training epochs (one point per epoch). A strong negative linear association is observed (Pearson $r = -0.985$, $R^2 = 0.970$), indicating that lower validation loss closely corresponds to higher ABPS during training.

monotonically (Pearson $r = -0.985$, $R^2 = 0.970$), revealing a clear linear relationship. This indicates that ABPS is not merely a mathematical construction, but a stable and sensitive indicator of the model's actual generalization gains. In particular, the consistent rise of ABPS alongside the gradual reduction in validation loss supports that ABPS captures broad ADMET performance improvements while remaining numerically well-behaved throughout training. The complete raw data for these ABPS calculations, including endpoint-wise RPPs and P_i values, are detailed in Table S1.

EXPERIMENT

All experiments were conducted on a single NVIDIA RTX A6000 GPU, and the model contains approximately 100 million parameters. We used a batch size of 32 and fixed the focal MAE hyperparameter at $\gamma = 2.0$ for all runs to ensure consistent experimental conditions. Under this setup, each training epoch required approximately 5 h. The maximum number of training epochs was set to 20. This upper bound was chosen to allow sufficient convergence under the scaffold-based 5-fold split setting, while also enabling us to monitor potential overfitting and representational instability that may arise during prolonged fine-tuning. In practice, validation performance tended to stabilize before 20 epochs in most cases, and no systematic overfitting pattern was observed within this range. The final model for each fold was selected based on ABPS, so that the chosen epoch maximized balanced performance across ADMET endpoints, rather than achieving a single peak score on a specific task.

To assess whether the proposed training strategy simultaneously achieves representational stability and broad ADMET performance gains, we analyzed the epoch-wise trajectories of validation loss, MLM accuracy, and ABPS (Figure 5). As shown in the figure, validation loss gradually decreased over training, and ABPS exhibited a corresponding monotonic increase. Meanwhile, MLM accuracy remained relatively stable without abrupt collapse, suggesting that the pretrained encoder's ability to capture SMILES structural and syntactic regularities is preserved during ADMET-driven fine-tuning.

We then compared the performance of the selected model against prior models reported on the TDC leaderboard. For each of the 22 ADMET endpoints, we computed performance using the official evaluation metric and summarized the results in Table 1. Our model exceeded the previously reported state of the art on 12 endpoints and achieved top-5 performance on 7 endpoints.

Despite these improvements, our model ranked relatively lower on Bioavailability, VDss, and Half-life compared with other endpoints. Unlike properties that map more directly to molecular structure—such as CYP reactivity, hERG binding, or metabolic clearance—these three endpoints reflect systemic pharmacokinetic processes that integrate absorption, distribution, metabolism, and excretion at the organism level.[23,24] Such characteristics are inherently difficult to fully explain using structure-only representations derived from SMILES. Nevertheless, the model maintained competitive performance within the range of the TDC leaderboard on these tasks and did not exhibit extreme degradation on any single endpoint. This indicates that the proposed encoder is particularly strong for endpoints with more direct structure–property correspondence, while still maintaining stable predictive capability for systemic kinetic properties without performance collapse.

Figure 6a visualizes the endpoint-wise relative performance profiles of different models. By normalizing each endpoint's performance to its relative position between a SOTA reference and a baseline (ref) model, this plot enables an intuitive comparison of how performance is distributed across tasks. Figure 6b reports the corresponding ABPS values for the same set of models. The comparison set includes representative GNN-based baselines known to be strong in ADMET prediction, such as MapLight+GNN,[25] Chemprop (D-MPNN),[26] AttentiveFP,[27] GIN (MiniMol),[28] and GCN.[29] For MapLight+GNN, Chemprop (D-MPNN), and GIN (MiniMol), we used the scores reported for each ADMET endpoint in a single-task setting. In contrast, AttentiveFP and our model produce all 22 endpoint predictions from a single encoder under a multi-task learning setup. While the inherent differences in learning setups (single-task vs. multi-task) suggest caution when interpreting direct performance gaps, we include these baselines to demonstrate that our single architecture can achieve performance levels comparable to a set of single-task models each specialized for an individual endpoint.

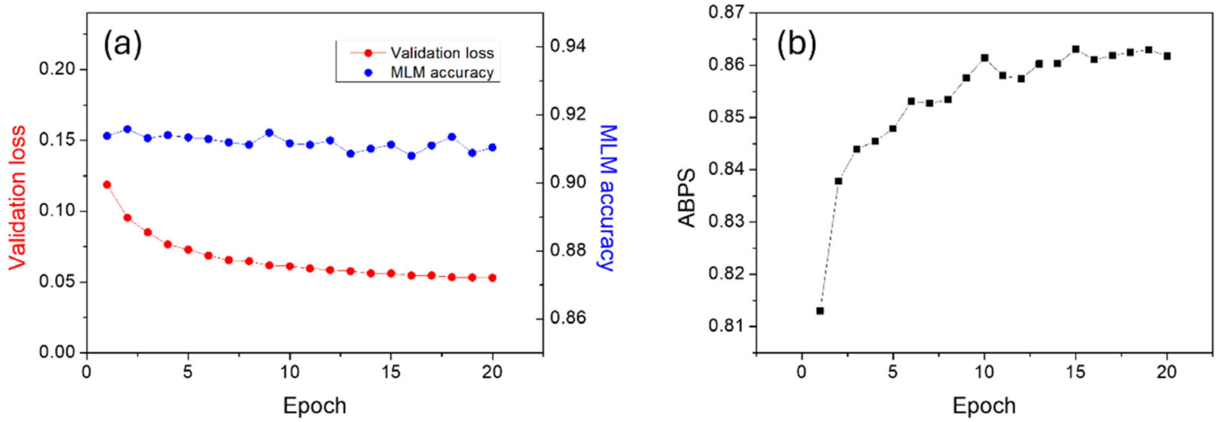


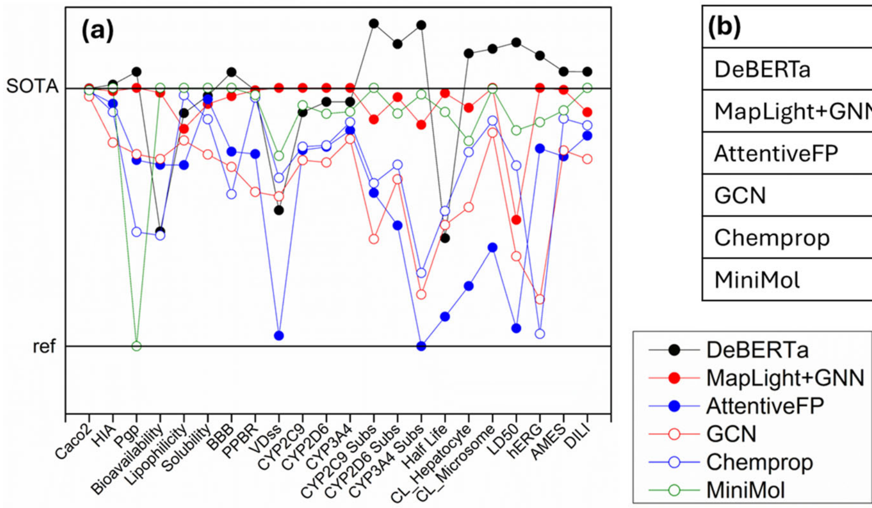
FIGURE 5 Training dynamics during multi-task ADMET fine-tuning. (a) Validation loss and MLM accuracy across epochs, showing steady loss reduction with stable MLM accuracy. (b) ABPS across epochs, demonstrating a monotonic increase over training.

TABLE 1 Performance of the proposed DeBERTa-based SMILES encoder on the TDC ADMET benchmark.

	TDC dataset		Leaderboard	DeBERTa-ADMET	
	Name	Metric	SOTA Result	Results	Rank
Absorption	Caco2	MAE(↓)	0.256 ± 0.006	0.271 ± 0.014	2
	HIA	AUROC(↑)	0.993 ± 0.005	0.998 ± 0.001	1
	Pgp	AUROC(↑)	0.938 ± 0.002	0.953 ± 0.003	1
	Bioavailability	AUROC(↑)	0.942 ± 0.002	0.862 ± 0.005	18
	Lipophilicity	MAE(↓)	0.456 ± 0.008	0.501 ± 0.012	5
	Solubility	MAE(↓)	0.741 ± 0.013	0.765 ± 0.010	3
Distribution	BBB	AUROC(↑)	0.924 ± 0.003	0.948 ± 0.002	1
	PPBR	MAE(↓)	7.440 ± 0.024	7.532 ± 0.117	3
	VDss	Spearman(↑)	0.713 ± 0.007	0.433 ± 0.023	16
Metabolism	CYP2C9	AUPRC(↑)	0.859 ± 0.001	0.810 ± 0.005	5
	CYP2D6	AUPRC(↑)	0.790 ± 0.001	0.750 ± 0.005	2
	CYP3A4	AUPRC(↑)	0.916 ± 0.000	0.892 ± 0.004	4
	CYP2C9 Substrate	AUPRC(↑)	0.474 ± 0.025	0.638 ± 0.005	1
	CYP2D6 Substrate	AUPRC(↑)	0.736 ± 0.024	0.846 ± 0.005	1
	CYP3A4 Substrate	AUROC(↑)	0.667 ± 0.019	0.740 ± 0.005	1
Excretion	Half Life	Spearman(↑)	0.576 ± 0.025	0.215 ± 0.020	15
	Clearance Hepatocyte	Spearman(↑)	0.536 ± 0.020	0.628 ± 0.007	1
	Clearance Microsome	Spearman(↑)	0.630 ± 0.010	0.767 ± 0.006	1
Toxicity	LD50	MAE(↓)	0.552 ± 0.009	0.492 ± 0.003	1
	hERG	AUROC(↑)	0.880 ± 0.002	0.927 ± 0.004	1
	AMES	AUROC(↑)	0.871 ± 0.002	0.891 ± 0.003	1
	DILI	AUROC(↑)	0.956 ± 0.006	0.988 ± 0.001	1
TDC leaderboard mean rank				3.8	

Note: For each of the 22 endpoints, the official evaluation metric is reported together with the corresponding leaderboard best (SOTA) score and the performance of our model. The rank denotes the model's position on the TDC leaderboard for each endpoint, and the mean rank across all endpoints is reported at the bottom. Corresponding citations for the SOTA models used for each endpoint are provided in Table S2. Red values indicate the top-ranked result and blue values indicate the 2nd–5th ranked results for each metric.

As shown in Figure 6b, the proposed model achieved the highest ABPS among all compared methods. This demonstrates that our SMILES-based Transformer is competitive not only with other SMILES encoders but also with graph-based GNN models that directly exploit molecular graph structure. The endpoint-wise distributions in Figure 6a further show that our gains are not driven by a small subset of tasks; rather, the model tends to match or surpass the SOTA reference across many endpoints and exhibits broadly and evenly distributed performance.



(b)	ABPS
DeBERTa	0.863
MapLight+GNN	0.856
AttentiveFP	0.705
GCN	0.733
Chemprop	0.756
MiniMol	0.828

FIGURE 6 Model comparison across the 22 ADMET endpoints. (a) Endpoint-wise tanh-normalized relative performance score (P_i), scaled between the SOTA reference (top horizontal line) and the lowest-ranked reference model (bottom horizontal line) to enable direct comparison across tasks. The proposed DeBERTa-based SMILES encoder is compared with representative GNN baselines (MapLight + GNN, Chemprop/D-MPNN, AttentiveFP, and GIN/MiniMol). (b) ABPS computed over all endpoints for each model; higher values indicate more balanced overall performance.

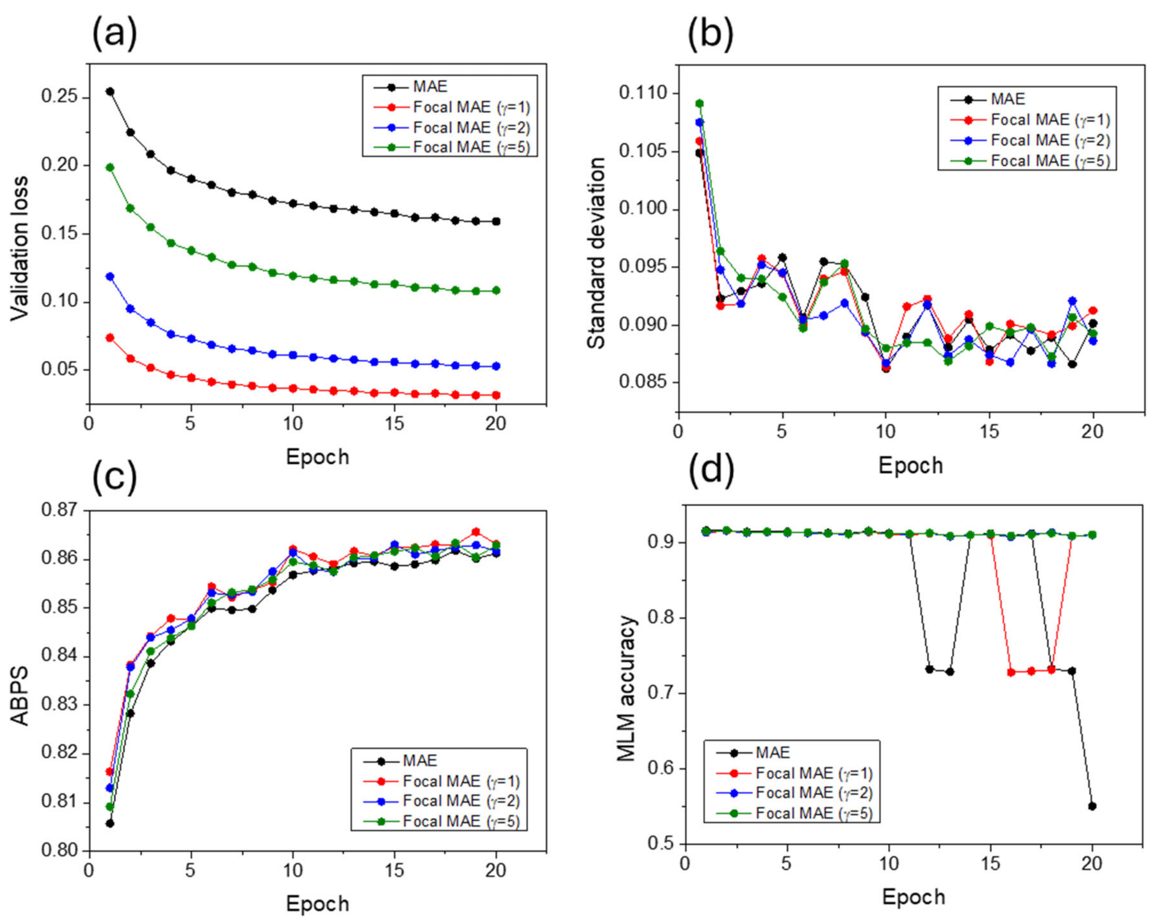


FIGURE 7 Loss-function ablation in multi-task ADMET fine-tuning. (a) Validation loss for MAE and focal MAE ($\gamma = 1, 2, 5$). (b) Epoch-wise standard deviation of the normalized endpoint scores (P_i) across the 22 ADMET endpoints. (c) ABPS across epochs under each loss formulation. (d) MLM accuracy during fine-tuning, highlighting differences in the stability of pretrained SMILES knowledge across loss functions.

Taken together, these results suggest that the model learns representations that more consistently capture ADMET-relevant structure–property relationships across endpoints, rather than optimizing narrowly for individual tasks. In particular, the strong performance relative to graph-based approaches indicates that a properly

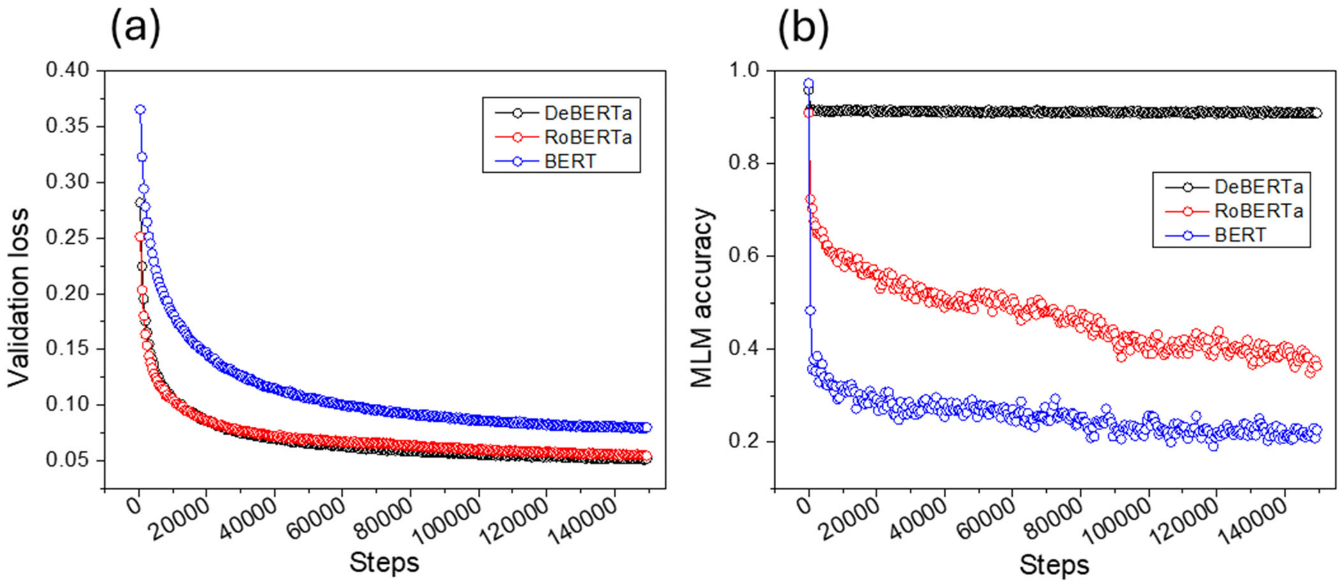


FIGURE 8 Comparison of encoder backbones under identical multi-task ADMET fine-tuning settings (focal MAE, $\gamma = 2$; learning rate $= 3 \times 10^{-5}$). (a) Validation loss for BERT-base, RoBERTa-base, and DeBERTa-base, showing similar convergence behavior. (b) MLM accuracy during fine-tuning, demonstrating substantially stronger retention of pretrained SMILES knowledge for DeBERTa relative to BERT and RoBERTa.

pretrained and stability-monitored SMILES encoder can achieve substantial chemical representational power even without explicitly encoding molecular graphs.

ABLATION STUDY

To examine how loss-function design influences the preservation of pretrained SMILES "language" competence during multi-regression ADMET fine-tuning, we conducted an ablation study comparing standard MAE with Focal MAE using $\gamma = 1$, 2 and 5 (Figure 7). Overall, ABPS did not show a meaningful difference across settings, and the standard deviation of endpoint-wise normalized relative performance (P_i) likewise converged to similar levels. Although the absolute magnitude of validation loss differed across conditions, this difference is best interpreted as a consequence of loss-definition–dependent scaling and did not exhibit a one-to-one correspondence with metrics that more directly reflect generalization performance.

In contrast, MLM accuracy revealed a clear dependence on the choice of loss function (Figure 7d). Under standard MAE and Focal MAE with $\gamma = 1$, MLM accuracy showed unstable oscillatory behavior, characterized by sharp drops followed by recovery over certain epoch intervals. This pattern indicates that optimization signals arising from ADMET regression can transiently and substantially reconfigure the shared encoder's representation space, momentarily weakening the model's ability to predict SMILES grammatical and structural regularities learned during pretraining. By comparison, for $\gamma = 2$ and $\gamma = 5$, such abrupt collapses were not observed and MLM accuracy remained relatively stable throughout training. These results suggest that error-adaptive reweighting in Focal MAE may reduce the extent to which optimization is dominated by a subset of endpoints or large-error samples, thereby mitigating abrupt representational perturbations and improving stability.

Following the loss-function analysis, we evaluated how the choice of architecture affects performance and representational stability during ADMET fine-tuning. We compared BERT-base (initialized from unikei/bert-base-smiles[30]), RoBERTa-base (initialized from entropy/roberta_zinc_480m[31]), and DeBERTa-base—models with broadly comparable parameter counts (BERT and RoBERTa have ~8.3 M and ~1.5 M more parameters than DeBERTa, respectively)—under identical training conditions (Focal MAE with $\gamma = 2$, learning rate 3×10^{-5}, and the same MTL setup) (Figure 8).

In terms of validation loss, all three models exhibited similar convergence behavior and reached comparable final values, indicating no pronounced architecture-dependent gap in ADMET regression predictive performance per se. However, clear differences emerged in MLM accuracy. For RoBERTa and BERT, MLM accuracy decreased steadily as fine-tuning progressed; this effect was especially pronounced for BERT, where the SMILES grammar-prediction capability established during pretraining weakened rapidly. In contrast, DeBERTa maintained MLM accuracy at a relatively higher level under the same conditions.

This disparity may be related to a structural difference in how these models encode content and positional information. BERT and RoBERTa combine token content and positional information into a single vector at the input stage, whereas DeBERTa factorizes content and position and models them through separated pathways in the attention mechanism. Such a factorized design may help distribute the impact of optimization signals arising from multi-task regression more locally within the representation space, which is consistent with the observed experimental outcome that DeBERTa preserves SMILES grammatical and structural competence more stably during fine-tuning.

LIMITATION

In this study, all molecules were represented using canonical SMILES to preserve a one-to-one correspondence between molecular graphs and input strings. As a result, input variability arising from alternative SMILES enumerations for the same molecule (e.g., randomized or non-canonical SMILES) was not included in training or evaluation. Systematic analysis of invariance and consistency across different SMILES representations is an important topic for understanding the generalization properties of SMILES-based models, and we leave this direction for future work.

CONCLUSION

This study aimed not to simply maximize ADMET prediction performance, but rather to develop a SMILES encoder that can effectively internalize ADMET-relevant structure–property information. To this end, we fine-tuned a DeBERTa-based molecular language model in a multi-task setting across 22 ADMET properties, while simultaneously analyzing how well the SMILES syntax and structural understanding acquired during pretraining are preserved throughout fine-tuning. In addition, recognizing that many ADMET endpoints differ in evaluation metrics and scales, we proposed ABPS as an integrated performance score to consistently assess the overall level of representation learning across tasks.

The proposed encoder achieved top-tier performance on many of the 22 endpoints in the TDC benchmark and demonstrated consistently competitive results under ABPS compared with baseline models. Notably, it showed comparable or superior performance even relative to GNN-based models, indicating that a properly pretrained and stably fine-tuned SMILES-based Transformer can attain ADMET-relevant representational capacity on par with graph-based approaches.

We also quantitatively examined the extent to which pretrained chemical "grammar" understanding is retained during fine-tuning. While the focal MAE–based loss design did not lead to large differences in final prediction accuracy, it mitigated the sharp decline in MLM accuracy, thereby reducing excessive degradation of SMILES token prediction capability. Furthermore, comparisons with BERT and RoBERTa suggested that DeBERTa's disentangled attention tends to preserve the grammatical and structural rules of SMILES more robustly in a multi-task ADMET training environment.

Taken together, this work demonstrates the feasibility of a SMILES encoder that incorporates ADMET information while preserving pretrained chemical syntax and structural knowledge. Such an encoder can serve as a foundation for more reliable molecular representations in future multimodal drug-design models, and it highlights that, when transferring NLP-based Transformer architectures to chemoinformatics, one should consider not only predictive performance but also the extent to which pretrained structural knowledge is maintained.

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DATA AVAILABILITY STATEMENT

The code used for model training, evaluation, and metric computation (including ABPS) is available at: https://github.com/aidanbio/AIdanChem. The processed datasets and scripts required to reproduce the experiments are provided in the same repository. The data that support the findings of this study are openly available in Google Drive at https://drive.google.com/drive/folders/1YmDrDAioDM_o92VwLHdsJJIRb30usbLN?usp=sharing.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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